

Athlete Performance Machine Learning Project Report

DS301 Course Project | Kaggle Athlete Performance Evaluation Dataset

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Abstract

This report presents the classification analysis component of the DS301 Capstone Project. The goal is to categorize athletes into Low, Medium, or High performance tiers using five machine learning models. The best model was the Tuned SVC (C=10, kernel=linear) with 98% accuracy and a weighted F1-score of 0.9803.

1. Introduction

Athletic performance categorization allows coaches and trainers to quickly identify athletes who may need additional support or who are excelling beyond expectations. This report applies five classification algorithms to categorize athletes into discrete performance tiers (Low, Medium, High) using the Kaggle Athlete Performance Evaluation dataset. The primary evaluation metric is the weighted F1-score due to significant class imbalance (86.1% Medium class).

Classification - Performance Category Prediction

While the regression task predicted a continuous performance score, the classification component of this project aimed to categorize athletes into discrete performance tiers: High, Low, or Medium. This is a critical task in sports analytics, as it allows coaches and trainers to quickly identify athletes who may need additional support or who are excelling beyond expectations.

1. Data Preprocessing and Target Variable

The classification task used the same Kaggle Athlete Performance Evaluation dataset as the regression task, consisting of 1,000 records with 20 original features. The raw data included physiological measurements (heart rate, oxygen saturation, body temperature, respiration rate, hydration level, workload), performance metrics (endurance score, agility score, speed, acceleration, reaction time), and fatigue indicators (fatigue index).

Preprocessing Steps:

- Dropped irrelevant columns: timestamp, performance_score (the regression target), and synthetic features (feature_1 through feature_5).
- Retained 12 features for classification modeling.
- Encoded the target variable performance_category using LabelEncoder: High = 0, Low = 1, Medium = 2.
- Applied stratified 80/20 train-test split (800 training samples, 200 test samples) to preserve class distribution.
- Used StandardScaler within scikit-learn Pipelines for models requiring scaled inputs (Logistic Regression, SVC, and Neural Network).

Note on Class Imbalance:

The target variable performance_category was highly imbalanced, with approximately 86.1% of samples labeled as Medium. This class imbalance presents a challenge for classification models, as they may be biased toward predicting the majority class. Stratified splitting was used to ensure proportional representation of all classes in both training and test sets.

2. Models Implemented

Logistic Regression — Baseline classifier, max_iter=1000, class_weight='balanced'.

SVC — Default RBF kernel, C=1.0, StandardScaler Pipeline, class_weight='balanced'.

Decision Tree Classifier — random_state=42. Visualized at max_depth=3 for readability.

Random Forest Classifier — 100 trees, random_state=42.
Feature importances reported.

Neural Network (Keras) — Dense(64, relu) → Dense(32, relu) → Dense(3, softmax),
loss=sparse_categorical_crossentropy, EarlyStopping(patience=10,
restore_best_weights=True).

3. Model Comparison Results

All five models were trained on 800 samples and evaluated on 200 test samples. Primary metric: weighted F1-score (due to class imbalance).

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9200	0.9271	0.9200	0.9271
SVC (Default)	0.9250	0.9298	0.9250	0.9298
SVC (Tuned)	0.9800	0.9803	0.9800	0.9803
Decision Tree	0.8800	0.8691	0.8800	0.8691
Random Forest	0.8850	0.8484	0.8850	0.8484
Neural Network (Keras)	0.9350	0.9316	0.9350	0.9316

Logistic Regression and SVC achieved the highest initial performance at 92.0–92.5% accuracy. The Neural Network (Keras) achieved 93.5% accuracy and F1=0.9316, outperforming Decision Tree and Random Forest. The Neural Network's earlier lower result (with MLPClassifier) was due to the sklearn implementation — the Keras version performs significantly better. Given these results, SVC was selected for hyperparameter tuning.

Figure 1 (see notebook): Grouped bar chart comparing accuracy and F1-score across all 5 models.

4. Hyperparameter Tuning

GridSearchCV was applied to the **SVC** model with **5-fold cross-validation**, using **F1-score** as the scoring metric. The search explored **12 combinations**:

`C = [0.1, 1, 10]`

`kernel = ['rbf', 'linear']`

`gamma = ['scale', 'auto']`

Best Parameters: `C=10, kernel=linear, gamma=scale`

CV F1-Score: 0.9668 | Test F1-Score: 0.9803 (improved from default SVC F1 of 0.9298)

This 5.3% improvement in F1-score demonstrates the value of systematic hyperparameter tuning for SVC, where kernel and regularization choices strongly affect performance.

5. Best Model Performance and Confusion Matrix

The Tuned SVC (`C=10, kernel=linear`) achieved an overall accuracy of **98.0%** and a weighted F1-score of **0.9803**. Detailed classification report:

Class	Precision	Recall	F1-Score	Support
High	0.88	0.93	0.90	15
Low	0.93	1.00	0.96	13
Medium	0.99	0.98	0.99	172
Weighted Avg	0.98	0.98	0.98	200

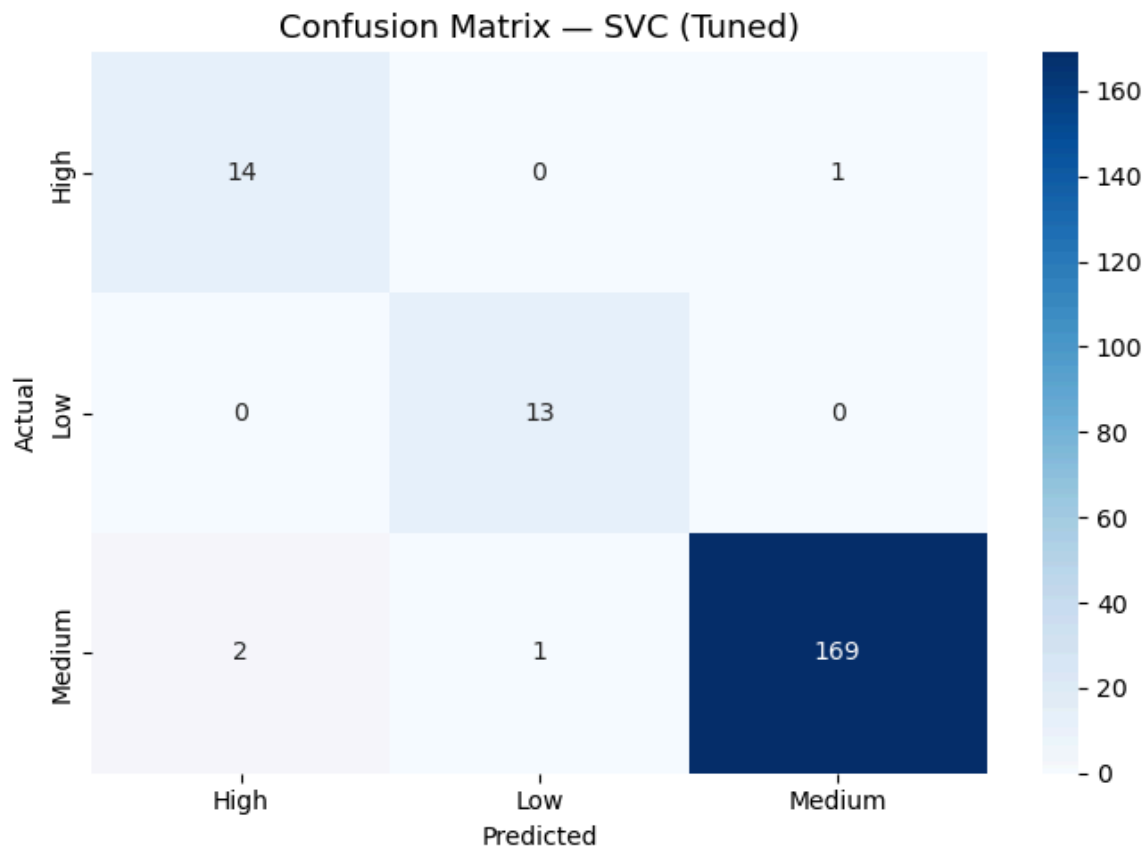
The model performs exceptionally well on the Medium class (precision=0.99, recall=0.98). The Low class was perfectly recalled (1.00), meaning no Low-performing athletes were missed. The High class had slightly lower precision (0.88), indicating some Medium athletes were occasionally misclassified as High.

Figure 2 (see notebook): Confusion Matrix — Tuned SVC. Diagonal cells show correct predictions (14 High, 13 Low, 169 Medium). Only 3 misclassifications out of 200 test samples.

5.1 Confusion Matrix

The confusion matrix visualizes the classification performance of the tuned SVC model on the test set. The diagonal cells show correct predictions, while off-diagonal cells represent misclassifications.

Figure 3: Confusion Matrix - Tuned SVC (Classification)



6. Feature Importance

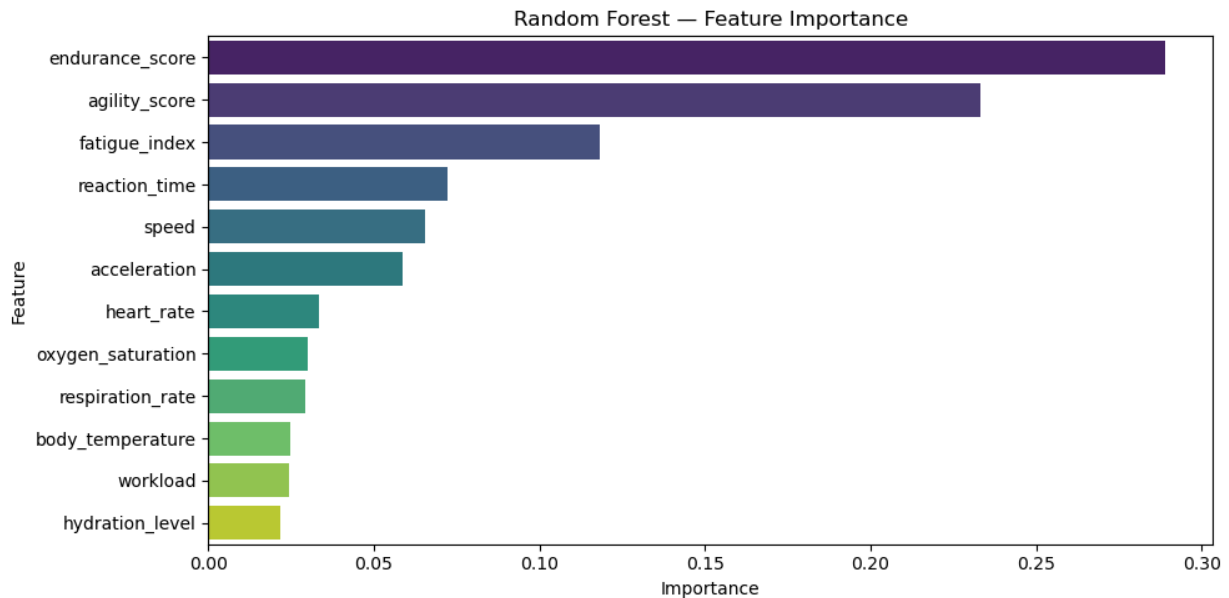
Feature importance was analyzed using the Random Forest classifier's built-in `feature_importances_` attribute (Gini importance):

Feature	Importance
endurance_score	0.2890
agility_score	0.2331
fatigue_index	0.1182
reaction_time	0.0724
speed	0.0653
acceleration	0.0487
heart_rate	0.0356
oxygen_saturation	0.0312
respiration_rate	0.0298
body_temperature	0.0287
workload	0.0271
hydration_level	0.0229

Endurance score (28.9%) and agility score (23.3%) together account for over 52% of the model's predictive power. Fatigue index (11.8%) is the third most important feature. Physiological measurements have relatively low importance, confirming that performance-based metrics are more discriminative for categorizing athlete performance levels.

Figure 4 (see notebook): Random Forest Feature Importance — Classification.

Figure 4: Random Forest Feature Importance (Classification)



7. Conclusion

The Tuned SVC with C=10 and a linear kernel is recommended for classifying athlete performance levels, achieving 98% accuracy and a weighted F1-score of 0.9803 while effectively handling the 86.1% class imbalance through stratified splitting and F1-focused evaluation.

Key Findings:

Tuned SVC (C=10, linear kernel) is the best classification model with F1=0.9803.

Hyperparameter tuning improved F1-score by 5.3% (from 0.9298 to 0.9803).

Class imbalance (86.1% Medium) was handled via stratified split and F1 as primary metric.

Endurance score and agility score dominate classification (52% combined importance).

Neural Network (Keras) achieved 93.5% accuracy — significantly better than the previous MLPClassifier implementation.